TIDES quick reference

Version 1.2 — harmonic prediction and the six coastal calculators.

Commands

`tides`	the menu
`tides find <text>`	search by name (every word must appear, any order)
`tides near <lat> <lon>`	stations nearest a position
`tides use <id>`	choose a station
`tides today [date]`	the day's table, curve, depth helper
`tides sky [date]`	the same, with sun and moon
`tides work <name> k=v …`	a calculator, working shown
`tides where`	the chosen station and the file locations
`tides day|night|plain`	colour mode

Search becomes a regular expression if it contains `^ $ . [ ] | ( ) * + ? { } \` — `"^st mary"`, `"bay$"`, `"falmouth|mystic"`.

The two signs that matter

Depth	charted depth **+** tide, against draught and clearance wanted
Air draught	charted clearance **−** tide, against your mast

Depths are below datum, so the tide adds. Clearances are above HAT, so the tide subtracts. Reversing that pair is how a mast meets a bridge.

Units, everywhere

angles	degrees true, 0–360
speeds	knots; heights and ranges, metres; distances, nautical miles
times	`HH:MM` or `HHMM`, **station standard time, no summer time**
positions	`lat,lon` decimal, **south and west negative**
variation, deviation	**east positive** (`var=-2` is 2° W)

A `+` in a URL is a space: write `tdhw2=5`, not `tdhw2=+5`.

The six calculators

name	what it gives	the parameters it must have
`twelfths`	height at a time: twelfths, cosine, and (at a station) harmonic	`lw lwh hw hwh t`, or `station t`
`secondary`	a secondary port from its standard port	the HW part, the LW part, or both
`stream`	a rate between neaps and springs	`spr nps range sprange nprange`
`atlas`	which atlas page, and where the range sits	`station` (+ `date t`)
`cts`	course to steer	`track stw streams`
`setdrift`	set and drift from a DR and a fix	`start co stw hours fix`

`streams` is `set:rate` per hour, comma separated — `135:1.5,150:2.0` is two hours. Vectors are summed and the triangle solved **once**, not hour by hour.

Leeway goes **into the wind**: `wind=port` means the wind is on the port side, so heading = water track − leeway.

Atlas pages run `HW -3` … `HW` … `HW +6`, each covering half an hour either side; **exactly half past goes to the later page**.

What it says when it will not guess

SECONDARY	no constants here; turns corrected from a reference, curve a half cosine
EXTRAPOLATED	beyond springs or neaps, or outside the almanac's limits
cannot make good this track	the stream beats the boat; the working shows why
means not available	not semidiurnal, or a secondary port — range given alone
one plain sentence naming a parameter	bad input; **nothing is worked**

Checked answers, for testing your own working

`twelfths lw=04:30 lwh=1.50 hw=10:52 hwh=4.39 t=07:00`	twelfths 2.48 m, cosine 2.47 m
`stream spr=4.2 nps=2.0 range=2.61 sprange=3.9 nprange=1.9`	2.78 kn
`cts track=255 stw=7.0 streams=030:1.5`	water track 246.3 T (steer 246), SOG 5.86 kn
the same `+ leeway=5 wind=port`	heading 241.3 T (steer 241)
`setdrift start=50.4630,-2.4550 co=250 stw=6.5 hours=2 fix=50.3660,-2.8260`	set 235 T, drift 2.4 nm, 1.2 kn

Rules of thumb

- The twelfths rule is a rule of thumb; where it and the harmonic height disagree, **the harmonic height is the better number**.
- Springs 2(M2 + S2), neaps 2(M2 − S2) — Admiralty, semidiurnal only.
- **Harmonics predict the astronomical tide and nothing else.** A deep low or a hard onshore blow moves the real water half a metre or more.
- Where this and the published tables, the almanac, the atlas or the chart differ, **they are right and this is wrong.**